

## Chapter 10: Determinants, the adjoint, and inverses

**Theorem:** If  $E$  is an elementary matrix and  $A$  is a square matrix of the same size, then  $\det(EA) = (\det E)(\det A)$ .

*Proof:* • If  $E \in \text{Type I}$ , then  $\det(EA) = -\det(A)$   
 and also  $\det(E) = -\det(I_n) = -1$ ,  $\Rightarrow \det(EA) = \det(E)\det(A)$   
 • If  $E \in \text{Type II}$ , then  $\det(EA) = k\det(A)$ ,  
 and  $\det(E) = k\det(I_n) = k$ , so  $\det(EA) = \det(E)\det(A)$ .  
 • If  $E \in \text{Type III}$ , then  $\det(EA) = \det(A)$ , and  $\det(E) = \det(I_n)$   
 so  $\det(EA) = \det(E)\det(A)$

**Theorem:** If  $A$  and  $B$  are  $n \times n$  matrices, then  $\det(AB) = \det(A)\det(B)$

*Sketch of proof (if time):*

- If  $A$  is invertible,  $A = E_1 \cdots E_k$ ,  $E_i$  are elementary matrices.  

$$\det(AB) = \det(E_k \cdots E_1 B) = \det(E_k) \cdots \det(E_1) \det(B)$$

$$= \det(E_k \cdots E_1) \det(B) = \det(A) \det(B)$$
- If  $A$  is not invertible,  $AB$  is also not invertible,  
 $\Rightarrow \det(A) = \det(AB) = 0 = \det(A) \cdot \det(B)$ .

**Example:** Compute the determinant of  $A$ ,  $B$ ,  $C$ ,  $AC$ , and  $BC$ , where  $A$ ,  $B$ , and  $C$  are the matrices

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 7 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \text{ and } C = \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}.$$

Confirm (if time) that  $\det(AC) = \det(A)\det(C)$  and  $\det(BC) = \det(B)\det(C)$ .

$$AC = \begin{bmatrix} 3 & 14 \\ -2 & 6 \end{bmatrix}, \det(AC) = 18 + 28 = 46$$

$$BC = \begin{bmatrix} 1 & 8 \\ 2 & 16 \end{bmatrix}, \det(BC) = 0.$$

$$\det(A)\det(C) = 23 \times 2 = 46$$

$$\det(B)\det(C) = 0 \times 2 = 0.$$

**Theorem:** If  $A$  is an  $n \times n$  matrix, then  $A$  is nonsingular if and only if  $\det(A) \neq 0$ . In this case,  $\det(A^{-1}) = \frac{1}{\det(A)}$ .

*Proof:*

• If  $A$  is invertible,  $\exists A^{-1}$ , s.t.  $A \cdot A^{-1} = I_n$

$$\Rightarrow \det(A \cdot A^{-1}) = \det(A) \cdot \det(A^{-1}) = \det(I_n)$$

$$\Rightarrow \det(A) \neq 0 \text{ \& } \det(A^{-1}) = \frac{1}{\det(A)} = 1.$$

• If  $\det(A) \neq 0$ , then the RREF of  $A$  also has nonzero determinant, since all 3 types of elementary row oper. cannot change a nonzero-determinant matrix to a zero-determinant one.

**Example:** Suppose  $A, B, C, D$  are  $3 \times 3$  matrices with  $\det(A) = 4, \det(B) = -3, \det(C) = 0, \det(D) = 6$ .

So the RREF of  $A$  has to be  $I_n$ .

a.) What's  $\det(A^2 B^{-1} D^T)$ ?

$$\begin{aligned} \det(A^2 B^{-1} D^T) &= [\det(A)]^2 \frac{1}{\det(B)} \cdot \det(D^T) \\ &= [\det(A)]^2 \frac{1}{\det(B)} \cdot \det(D) = 16 \times \frac{1}{-3} \times 6 = -32. \end{aligned}$$

b.) Is  $BC$  an invertible matrix?

$$\det(BC) = \det(B) \det(C) = 0, \text{ so NOT invertible.}$$

c.) If  $F$  is a  $3 \times 3$  matrix with  $F = P^{-1}BP$  for some invertible matrix  $P$ , what's  $\det(F)$ ?

$$\begin{aligned} \det F &= \det(P^{-1}BP) = \det(P^{-1}) \det(B) \det(P) \\ &= \frac{1}{\det(P)} \cdot \det(B) \cancel{\det(P)} = \det(B) \\ &= -3 \end{aligned}$$

**Example:** Let  $A$  be a  $4 \times 4$  matrix with  $\det(A) = 3$ .

(a) Describe the set of all solutions to the homogeneous system  $A\mathbf{x} = \mathbf{0}$ .

$\det(A) = 3 \Rightarrow A$  invertible  $\Rightarrow A\mathbf{x} = \mathbf{0}$  has only trivial sol'n.

(b) If  $A$  is transformed to reduced row echelon form  $B$ , what is  $B$ ?

$$B = I_4.$$

(c) Give an expression for a solution to the linear system  $A\mathbf{x} = \mathbf{b}$ , where

$$\mathbf{b} = \begin{bmatrix} 4 \\ 2 \\ 3 \\ 1 \end{bmatrix}.$$

$$\mathbf{x} = A^{-1} \cdot \begin{bmatrix} 4 \\ 2 \\ 3 \\ 1 \end{bmatrix}$$

(d) Can the linear system  $A\mathbf{x} = \mathbf{b}$  have more than one solution? Explain.

NO, only 1 sol'n, since the sol'n is  $A^{-1}\mathbf{b}$ .

(e) Does  $A^{-1}$  exist?

Yes.

### Invertibility Criteria

Let  $A$  be an  $n \times n$  matrix. The following are equivalent:

1.  $A$  is invertible/nonsingular.
2.  $A$  is row equivalent to  $I_n$ .
3.  $A\mathbf{x} = \mathbf{0}$  has only the trivial solution.
4. The linear system  $A\mathbf{x} = \mathbf{b}$  has a unique solution for every  $n \times 1$  matrix  $\mathbf{b}$ .
5. The linear system  $A\mathbf{x} = \mathbf{b}$  is consistent for every  $n \times 1$  matrix  $\mathbf{b}$ .
6.  $A$  is a product of elementary matrices.
7.  $\det(A) \neq 0$ .

# UNIT 2: VECTOR SPACES AND EIGENVALUES

## Chapter 12: Introduction to Vectors

---

*Scalars*: completely described by giving their magnitude. (Mass, pressure, temperature, etc.)

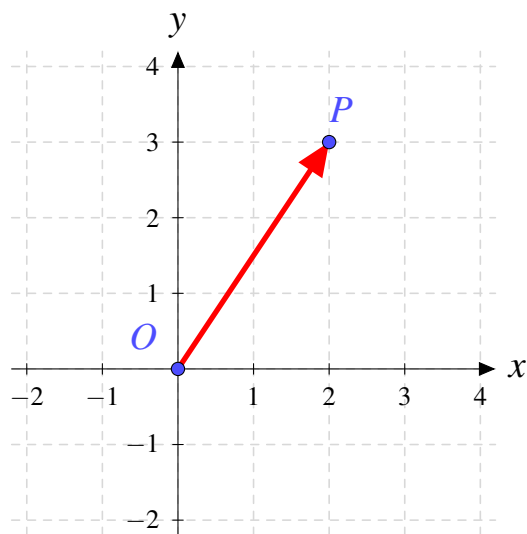
*Vectors*: require for their description not only a magnitude, but also a **direction**. (Velocity, acceleration, force, etc.)

### Vectors

With each **point**  $P$  in the plane, we can associate an **ordered pair** of real numbers, its **coordinates**. Conversely, we can associate a point in the plane with each ordered pair of real numbers.

A **vector** can be associated with a  $2 \times 1$  matrix  $\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$ . The initial point is the **origin** and the terminal point is  $P(x, y)$ . The set of all vectors in the plane is denoted by  $\mathbb{R}^2$ , it is also called **2-space**.

Note: Your homework represents vectors  $\begin{bmatrix} x \\ y \end{bmatrix}$  as  $\langle x, y \rangle$ .



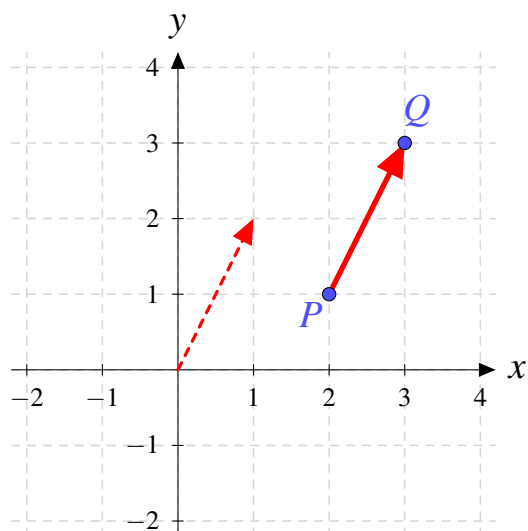
Notation:  $\overrightarrow{OP}$ ,  $O$  is the **tail**, and  $P$  is the **head**.

A vector has a **direction**, and a **magnitude**, which is the **length** of the line segment.

The vectors  $\mathbf{v} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$  and  $\mathbf{u} = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$  are **equal** if and only if  $x_1 = x_2$  and  $y_1 = y_2$ .

If the tail of a directed line segment is not the origin, but  $P(x_1, y_1)$ , and its head is  $Q(x_2, y_2)$ , then it can be represented by the vector

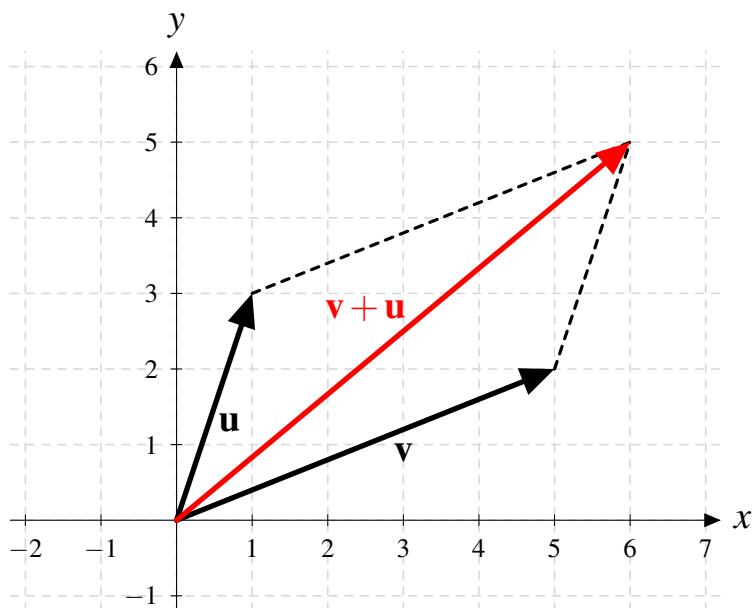
$$\overrightarrow{PQ} = \begin{bmatrix} x_2 - x_1 \\ y_2 - y_1 \end{bmatrix}.$$



*Definition:* Let  $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$  and  $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$  be two vectors. The **sum** of them is

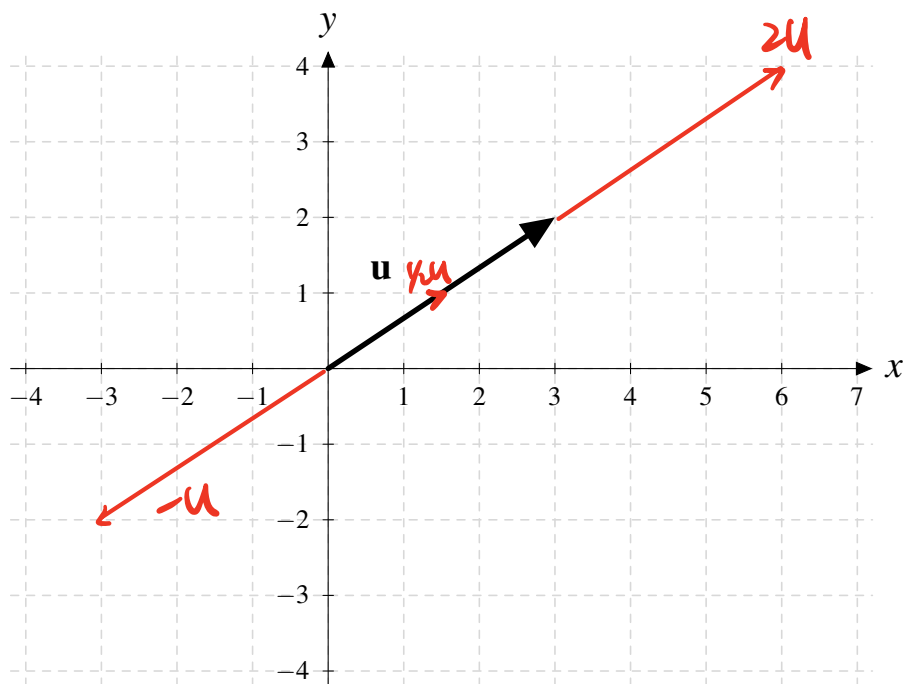
$$\mathbf{v} + \mathbf{u} = \begin{bmatrix} v_1 + u_1 \\ v_2 + u_2 \end{bmatrix}.$$

We can interpret vector addition geometrically:



*Definition:* If  $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$  is a vector, and  $c \in \mathbb{R}$  is a scalar, then the **scalar multiple**  $c\mathbf{v}$  is  $c\mathbf{v} = \begin{bmatrix} cv_1 \\ cv_2 \end{bmatrix}$ .

**Example:** For  $\mathbf{u}$  given below, draw  $2\mathbf{u}$ ,  $\frac{1}{2}\mathbf{u}$ , and  $-\mathbf{u}$ .



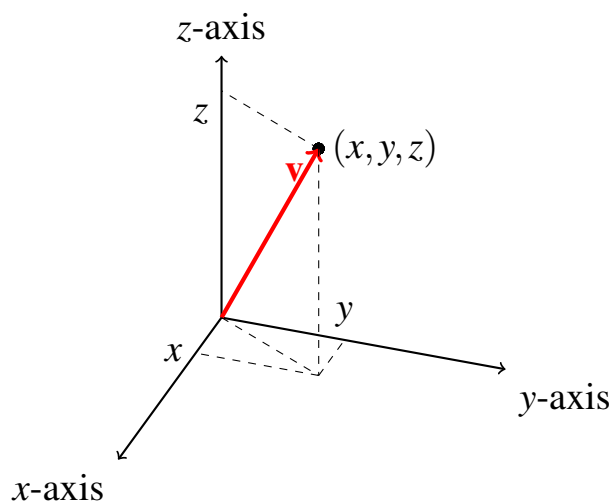
*Definition:* Two nonzero vectors are **parallel** if they are scalar multiples of each other.

### Vectors in 3-Space and $n$ -space

We can create vectors in higher dimensional space. In 3-space (or  $\mathbb{R}^3$ ), we have three coordinate axes, and the origin is the point (0,0,0).

With each point of the space, we can associate an ordered triple  $(x, y, z)$  of real numbers,

its **coordinates**. And we can associate a vector (or 3-vector)  $\mathbf{v} = (x, y, z) = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ .



Similarly to the plane, we can define addition, scalar multiplication in 3-space.

A vector in  $n$ -space (or in  $\mathbb{R}^n$ ) is

$$\mathbf{v} = (x_1, x_2, \dots, x_n) = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

where each  $x_i$  is a real number.

Similarly to  $\mathbb{R}^3$ , we can define addition, scalar multiplication in  $n$ -space via coordinate-wise addition/scalar multiplication.

*Definition:* If  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$  are vectors in  $\mathbb{R}^n$  and  $c_1, c_2, \dots, c_k$  are real numbers, then an expression of the form

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k$$

is called a **linear combination** of  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ . The scalars  $c_1, \dots, c_k$  are called **coefficients**.

Note: We've seen this definition already for general matrices.

**Example:** Which vectors in  $\mathbb{R}^2$  are linear combinations of  $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$  and  $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ ?

$$c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \vec{b} \Leftrightarrow \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \vec{b}$$

Since  $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$  is invertible, for each  $\vec{b} \in \mathbb{R}^2$ ,  $\exists$  a sol'n

$$\vec{c} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^{-1} \cdot \vec{b}, \text{ so all vectors in } \mathbb{R}^2 \text{ is a L.C. of } \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ \& } \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

**Example:** Which vectors in  $\mathbb{R}^3$  are linear combinations of  $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ ,  $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ , and  $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ ?

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ is invertible, similar to the above,}$$

all vectors in  $\mathbb{R}^3$  is a L.C. of  $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$  &  $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ .



*Definition:* Consider  $\mathbb{R}^n$ , and let  $\mathbf{e}_j$  denote the  $j$ th column of  $I_n$ , i.e.,

$$\mathbf{e}_j = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \text{--- } j^{\text{th}} \text{ row.}$$

These vectors are called the **standard basis vectors** for  $\mathbb{R}^n$  (this name will make sense later).

### Important properties of vectors

**Theorem:** If  $\mathbf{v}$ ,  $\mathbf{u}$  and  $\mathbf{w}$  are vectors in  $\mathbb{R}^n$ , and  $c$  and  $d$  are real numbers (scalars), then the following properties are valid:

1.  $\mathbf{v} + \mathbf{u} = \mathbf{u} + \mathbf{v}$
2.  $\mathbf{v} + (\mathbf{u} + \mathbf{w}) = (\mathbf{v} + \mathbf{u}) + \mathbf{w}$
3.  $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$
4.  $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$
5.  $c(\mathbf{v} + \mathbf{u}) = c\mathbf{v} + c\mathbf{u}$
6.  $(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$
7.  $c(d\mathbf{v}) = (cd)\mathbf{v}$
8.  $1\mathbf{v} = \mathbf{v}$

*Proofs of some of the properties for  $\mathbb{R}^2$ :*

proofs are the same to those in matrix operations.